

Statistical genetics on All of Us

Lab 1: your Workbench, your data — EDA, a phenotype, and a PGI

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September 2026

“Life is not invulnerable, but it is extraordinarily good at finding ways through adversity...”

— Philip Ball, *How Life Works* (2023)

20 minutes of instruction 40 minutes of practice

What you will do today – on real data

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- 1 **Explore:** summary statistics and histograms of a real variable
 - 2 **Construct a phenotype:** from raw CDR records to one clean, QC'd, plausibility-bounded row per person
 - 3 **Build a PGI:** PLINK on one chromosome, from posted GWAS summary statistics
 - 4 **Model:** a basic regression with the PGI in it
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The ground rules. Person-level files stay in the workspace, forever; printed outputs are aggregate and screened. No answer key – your numbers are real; the handout gives plausibility checks. Queries and copies cost money, capped in the scripts.

Confidence with your actual Workbench is the deliverable.

Where today's data live

Resource	What we take from it
CDR (BigQuery)	Height measurements + person table → our phenotype
SSGAC shared bucket	HapMap3-filtered PLINK fileset, one chromosome
v9 controlled bucket	Genetic-ancestry predictions (requester-pays: fetches must name a billing project)
Workshop bucket	Instructor-staged PGI weights; your saved runs

Cloud files must be **copied to the VM disk** before PLINK can touch them. Copies are fast inside Google's network – and counted, every time.

Paths are variables in `config.aou.sh` – which is why the same scripts also run against the local sandbox.

A phenotype is a chain of decisions

```
WHERE measurement_source_concept_id = 903133    -- not LOINC 3036277
```

- **Concept choice:** the program's own measurement concept. The standard LOINC concept “works” – and silently pools in hundreds of thousands of EHR heights.
- **One row per person:** we take each person's *median* (mean and first-visit are rival, defensible choices – pick one, state it).
- **Plausibility bounds:** 100–250 cm, with removals counted.
- **Standardize within sex;** “other” handled explicitly, not silently.
- Still pooled underneath: clinic and remote self-report share this concept.

Every line of the cleaning script is a methods sentence.

Real files, real conventions

File	Its quirk
<code>.fam</code> (SSGAC fileset)	Key spelled FID=IID; other AoU filesets use FID=0 – a mismatched keep-list matches zero samples, silently
<code>.bim</code>	rsIDs here; AoU ACAF callsets use variant IDs – crossing schemes needs the VAT map
<code>ancestry_preds.tsv</code>	Key column named <code>research_id</code> – same numbers as <code>person_id</code> , different name
<code>.pheno</code> files	Can contain literal NA: a value that crashes or poisons, depending on your parser

Read two lines of every file before joining it. Count every join.

Cleaning and EDA, with disclosure built in

- First look at the phenotype: N, mean, SD, quartiles – overall and by sex. Is the mean plausible for a US cohort? Is the SD? Would a spike or second mode implicate a concept or unit upstream?
- Printed counts are **rounded to the nearest 100**, and 1–20 never appears – so no table cell, or difference of cells, can disclose a small count. Exact Ns exist only in workspace-side files.
- Histogram bars are counts too; sparse bins can whisper. Figures get a human look before any dissemination.

The funnel stays narratable and disseminable at the same time.

Real weights, real match counts

```
p2 --bfile lab_data/chr22_filtered \  
  --score lab_data/height_weights.txt 1 2 3 header \  
      cols=+scoresums list-variants --out results/aou_pgi
```

- Weights come from **GWAS summary statistics posted to the shared USC pod bucket**, rsID-keyed to match this fileset. Your first read of the log is the match count.
- One chromosome $\Rightarrow$ a deliberately **partial** PGI. Real studies score all 22 and sum – same command, a loop, and patience.
- Missing genotypes are mean-imputed by default; state it.

A PGI is a weighted allele count. The work is the honest match.

Reading the ancestry figure honestly

The lab's last figure overlays PGI densities by AoU's predicted genetic ancestry.

- Distributional differences reflect **allele frequencies and the discovery GWAS behind the weights** as much as any biology. Say so wherever the figure appears.
- **Portability**: prediction accuracy is not constant across groups – a PGI built from mostly-European discovery GWAS predicts less well at greater ancestral distance. In AoU this determines who your index serves.
- Ancestry here is a statistical prediction from genotypes – not race, not ethnicity, not a quality score.

Two facts, always together: distributions differ, and accuracy differs.

A regression with the PGI in it

$$\text{height_}z_i = \alpha + \beta \text{PGI}_i + \gamma \text{age}_i + \delta \text{female}_i + \sum_{k=1}^5 \theta_k \text{PC}_{ik} + \varepsilon_i$$

- Both sides standardized, so β reads: *SD of height per SD of PGI*, conditional on the covariates.
- Report the PGI's **incremental** R^2 – what it adds beyond the covariates – not its R^2 alone.
- The PCs come from AoU's own ancestry file; leaving them out moves β , and that movement *is* structure.
- Honest labels: association, not cause; one chromosome, not a full index; independent-samples OLS, not a family model.

One command in the lab: `bash scripts/14_pgi_regression.sh`

Costs, time, and etiquette

- Queries are byte-capped ($\sim$ \$0.20 each at the default); bucket fetches bill the workshop project; today's whole lab is a few dollars per seat.
- Default cohort: a **1-in-10 teaching subset** (`SAMPLE_MOD=10`) – fast queries, quick joins. `SAMPLE_MOD=1` is the same code on everyone, for your real projects.
- Every script was rehearsed in a free local sandbox that mirrors AoU file conventions – take it home to debug your own code before paying for a VM. A green sandbox run proves plumbing, not science.
- **Stop your cloud app when done.** VMs bill while idle.

Cheap, capped, rehearsed – and turned off afterward.

Your 40-minute lab

- 20–26 Copy `config.aou.example.sh` → `config.aou.sh`; fill values; pre-flight.
 - 26–31 Stage the chromosome + ancestry file; build the phenotype from the CDR.
 - 31–36 Phenotype EDA: read the funnel, open both figures.
 - 36–44 Fetch weights, score, PGI EDA; open the ancestry figure.
 - 44–49 The regression: read the coefficient table with a partner.
 - 49–54 Interpretation questions.
 - 54–60 Save (the gate is live), **stop the app**, exit check.
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```
cp config.aou.example.sh config.aou.sh      # then edit its values
export WORKSHOP_CONFIG=config.aou.sh
bash scripts/00_preflight.sh
bash scripts/10_fetch_aou_genotypes.sh && bash scripts/11_build_phenotype.sh
bash scripts/12_pheno_eda.sh && bash scripts/13_score_real_pgi.sh
bash scripts/14_pgi_regression.sh && bash scripts/04_save.sh
```

Questions to answer while your code runs

1. Which allele-and-ID conventions did today's three joins have to survive?
2. Why the median per person? Name one alternative and its trade-off.
3. Your funnel prints ~59,300. Why not the exact number?
4. What match rate did `-score` report, and what does a low one mean?
5. Give the two-sentence honest reading of the ancestry figure.
6. In the regression, why report incremental R^2 rather than R^2 alone – and what happens to β if the PCs come out?
7. What would change in the code to run everyone instead of 1-in-10? To score all 22 chromosomes?

The exit check

“I pulled real records, defended every cleaning step, counted every join, and put a real PGI in a model I can defend.”

For your own project

Swap the concept ID, the bounds, and the one-row rule for your phenotype – the chain is the same.

Swap the staged weights for your trait's release, and score all 22 chromosomes.

Keep the defaults that protected you today: caps, counts, rounding, the gate, and the sandbox first.

“... the astonishing process that created a form of matter able to begin understanding itself: us.”

— Philip Ball, *How Life Works* (2023)

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